

Sound, space & *perception.*

Sound artist · Physicist · Researcher
Buenos Aires, Argentina

[Vimeo ↗](#)

[YouTube ↗](#)

[Website ↗](#)



Sonic Crystal Room · Manuel Eguía / Oscar Edelstein

A practice grounded in research. A life working with artists.

My professional foundation is research in physics, nonlinear dynamics, neuroscience and acoustics. For 25 years I have worked at an art school, training artists and forging connections between art and science through laboratory practice and international collaborative projects. From that sustained exchange, I have developed a distinct artistic practice.

I explore emerging questions in sound, acoustics and auditory perception by making works in which those questions become experiences. A room can become an instrument; a familiar interior can unfold into an unfamiliar world; a landscape can be heard through the relationship between a voice and the space that answers it.

My work moves between physical acoustics, immersive sound, virtual reality and collective listening. Scientific models, experiments and technical development are part of how I make art. The encounter with a listener, a body and a particular place gives that work its form.



Manuel Eguía is an artist and physicist based in Buenos Aires. He holds a PhD in Physics from the Universidad de Buenos Aires (2002), is a CONICET researcher and a full professor at the Escuela Universitaria de Artes, Universidad Nacional de Quilmes. He founded LAPSo, the Laboratory of Acoustics and Sound Perception, in 2007, and founded Proyecto GRAPa in 2017. Since 2026, he directs UNQ's Instituto de Investigaciones en Arte, Ciencia, Tecnología y Sociedad (ArCiTeS).

Sonic Crystal Room

A room that becomes an instrument

Manuel Eguía / Oscar Edelstein



Acoustic installation · Performance environment

[Watch the work ↗](#)

An architecture of mobile modules and rotating wooden columns transforms sound through physical acoustics. The room itself becomes a musical instrument.

Sonic Crystal Room makes the acoustic behaviour of a space available to composition and performance. Its columns form a sonic crystal: a repeated structure that changes how sound travels. Reconfiguring that structure changes what a listener hears and where the sound seems to be.

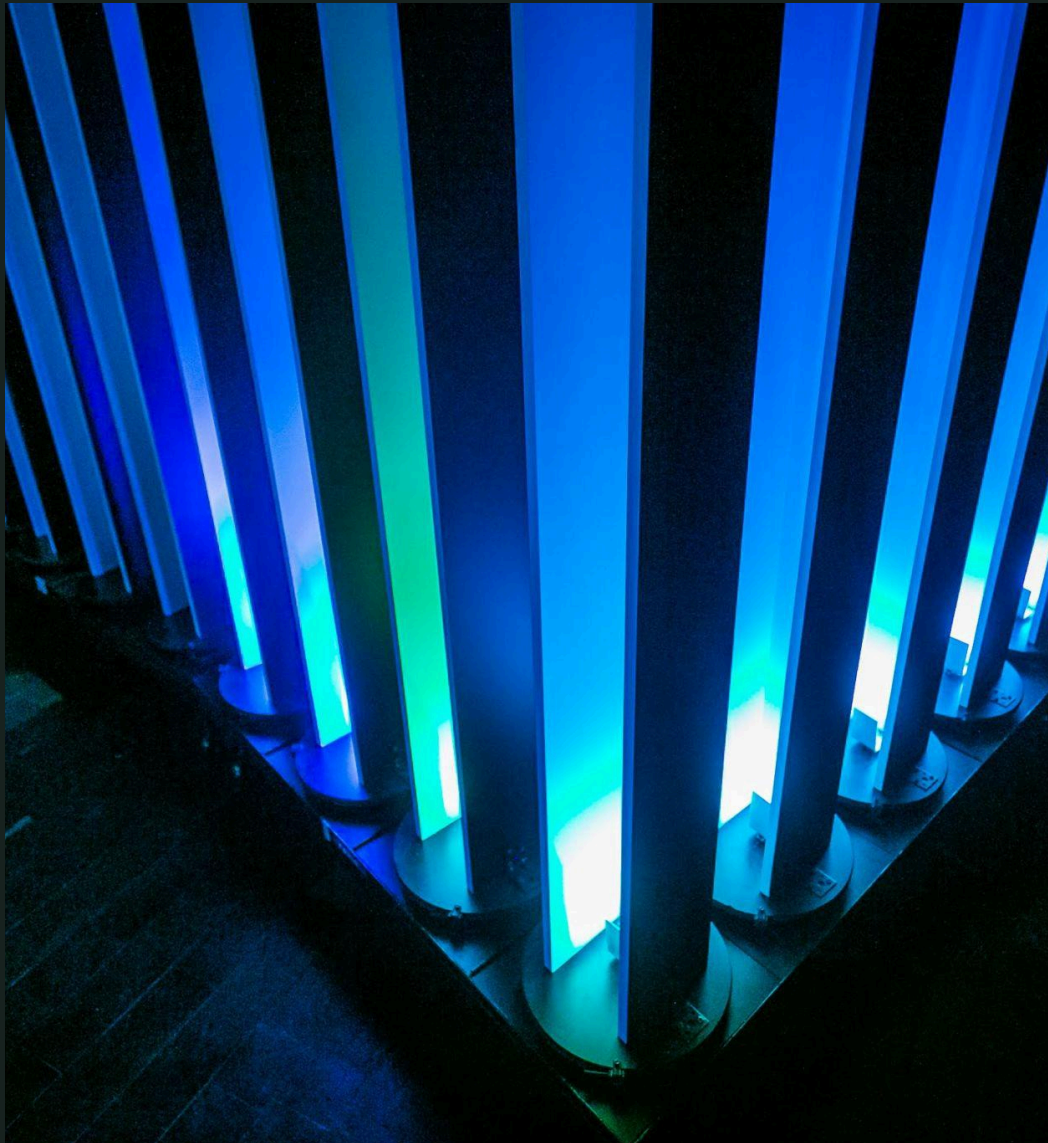
The work makes architecture an active part of a performance. Movement, apparent distance and shifts in acoustic perspective emerge from the relationship between sound sources, the columns and the listener.

MY ROLE

Co-creator with Oscar Edelstein; acoustic research and development.

CONTEXT

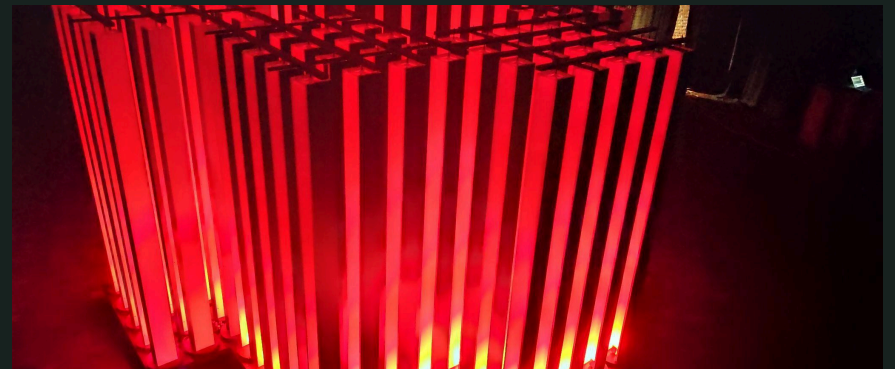
First presented at UNQ in 2019. Presented in the opening programme of the Festival de Nueva Ópera, cheLA, Buenos Aires, September 2024.



Composing with the space itself

Mobile modules and rotating columns create a room whose acoustic properties can change during a performance. Focusing, redirection and scattering become compositional resources, alongside the sound produced by musicians or loudspeakers.

Outside performances, the installation can be explored as a three-dimensional score. Visitors move through an architecture whose geometry and material properties organize the listening experience.



Mobile structures, changing acoustic perspectives.
Manuel Eguía / Oscar Edelstein

Transit

Inside, outside, elsewhere

Manuel Eguía / Mauro Zannoli



Site-specific virtual reality installation

[Watch the virtual journey ↗](#)

[Northern Sustainable Futures ↗](#)

A virtual journey begins in an exact replica of the room the participant occupies. Familiar space gradually opens into impossible interiors and transformed landscapes.

Transit is conceived as a series of site-specific virtual reality experiences. Each episode starts from the participant's immediate physical surroundings, making the transition into a virtual world difficult to locate. Local architecture, landscape and cultural references become the material of the work.

The first episode inhabits iM Konsthall, a mobile gallery built inside a shipping container. Its interior multiplies and transforms through a sequence of spatial folds, until the participant finds themselves outside, in the northern Swedish forest at night, looking for the way back.

MY ROLE

Co-creator with Mauro Zannoli.

CONTEXT

Commissioned by Northern Sustainable Futures for iM Konsthall. Developed during an artist residency in Moskosel, Sweden, October 2023.

Returning virtual space to a place

The container is both the physical venue and an image of transit. Its windows initially frame the same landscape seen before putting on the headset. That continuity provides the perceptual anchor from which the installation departs.

Transit brings territory and context into digital experience. The journey returns to its starting point, but the passage through altered spaces changes the participant's relationship to that familiar room.

*The same room.
A different sense of where we are.*

First episode: iM Konsthall, Moskosel, Sweden, 2023.

Manuel Eguía / Mauro Zannoli

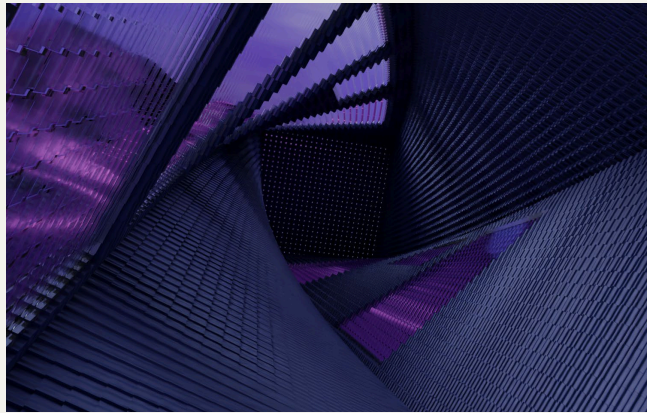
Commission: Northern Sustainable Futures

[Watch the virtual journey ↗](#)

[Northern Sustainable Futures ↗](#)



01 / A familiar room



02 / Spatial transformation



03 / Returning from outside

Proyecto GRAPa

Listening to a voice and the place that answers it

Proyecto GRAPa · Manuel Eguía, Francisco Durante, Damián Payo, Mauro Zannoli



Acoustic heritage · Immersive recording · Documentary

[Watch the copla recording · 360° ↗](#)

An enquiry into acoustic heritage through the relationship between copla singing, its communities and the natural spaces in which it is practised.

GRAPa — Grupo de Relevamiento Acústico del Patrimonio — brings together acoustic measurement, immersive audiovisual recording and documentary filmmaking. I founded the project to investigate and document the relationship between musical practices and their acoustic environments.

La Copla y el Anfiteatro de la Quebrada de las Conchas centres on a natural amphitheatre in the Calchaquí Valleys. With the active participation of coplera Mariana Carrizo and local singers, the project records the interplay of voices, cajas and rock walls. The acoustic character of the place is part of the musical experience.

MY ROLE

Founder and member of the research and production team.

CONTEXT

Formed in 2017 through cooperation between UNQ and Universidad Católica de Salta. Fieldwork at the natural amphitheatre of Quebrada de las Conchas, Salta, April 2018. Latin GRAMMY Cultural Foundation research support, 2021.

Three ways of documenting a listening culture



01 / IMMERSIVE RECORDING

Immersive copla recordings place the listener within the singing circle, preserving the spatial relationships between performers. The work is grounded in the knowledge and participation of the singers, including Mariana Carrizo.

02 / ACOUSTIC SURVEY

Ambisonic and binaural impulse-response measurements capture how the amphitheatre responds to sound. These recordings allow aspects of its acoustic character to be recreated through loudspeakers or headphones.

03 / DOCUMENTARY

A documentary follows the April 2018 fieldwork, bringing together the acoustic survey and the recording of coplas. Together, the three strands present music and the space of its creation as a shared acoustic heritage.

Founder: Manuel Eguía

Team: Francisco Durante, Damián Payo, Mauro Zannoli and Manuel Eguía.

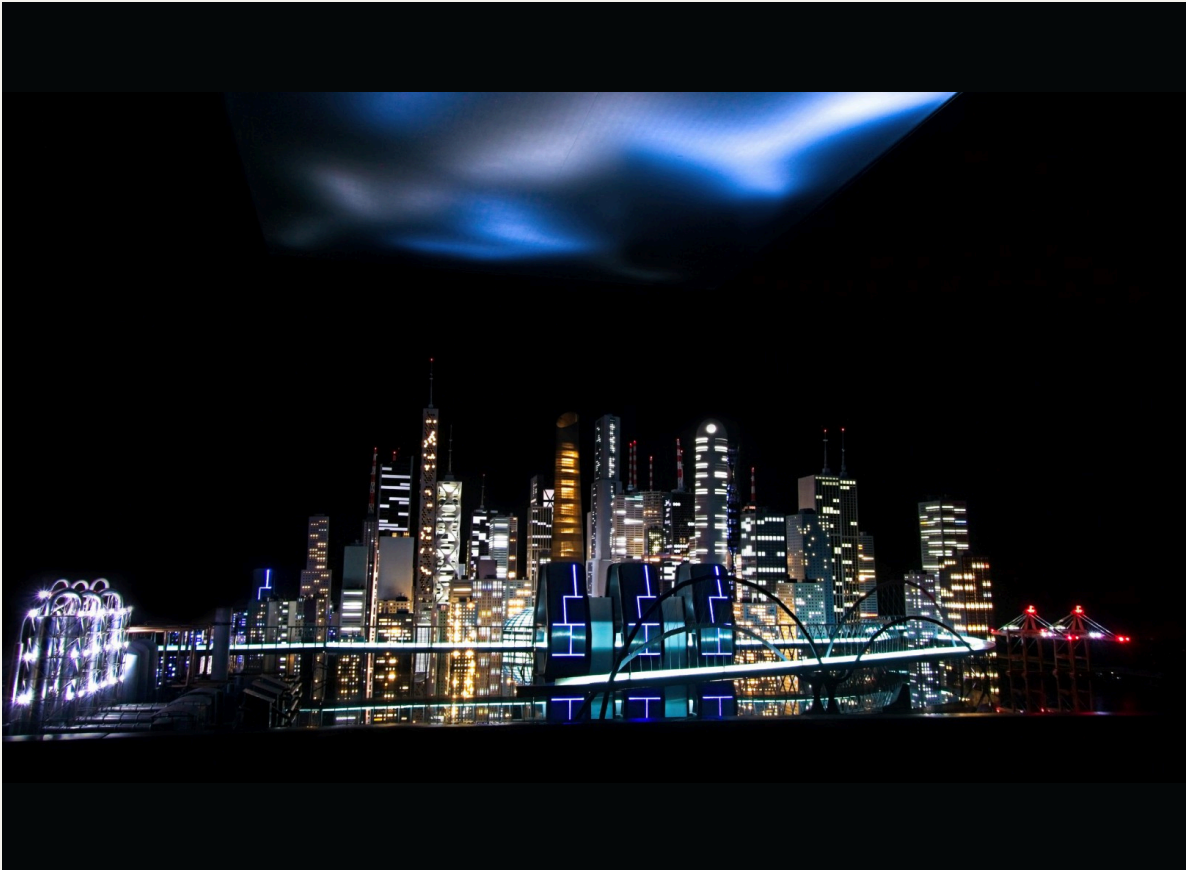


La Copla y el Anfiteatro de la Quebrada de las Conchas · Salta, Argentina
GRAPa, with Mariana Carrizo and local copleras and copleros.

Biocenosis

A city as an interdependent system

Sigismond de Vajay & Kevin Lesquenner + LAPSo



Interactive installation · Sculpture, light and sound

[Watch the installation ↗](#)

[Explore the VR version ↗](#)

An immersive city model stages the interdependence of human, urban and non-human life under environmental pressure.

A city of 430 buildings becomes the setting for an encounter with ecological fragility. Sculpture, lighting and interactive systems form a submerged urban landscape, in which visitors' presence modulates the behaviour of the fictional city.

Developed by Sigismond de Vajay in collaboration with Kevin Lesquenner and LAPSo, the installation connects artistic world-building with systems inspired by biological processes. My participation formed part of LAPSo's collective contribution to the work.

MY ROLE

Collaborating artist and researcher within LAPSo. Biennial team: Manuel Eguía, Lucas González, Pablo Riera and Sol Gómez.

CONTEXT

13th Mercosul Biennial, Porto Alegre, Brazil, 15 September–20 November 2022.

The laboratory, the classroom, the work

At LAPSo, research and artistic production develop through sustained collaboration. The laboratory brings together acoustic experimentation, sound spatialization, instrument design, auditory perception and virtual environments. It is also a place where artists develop work and researchers learn from artistic questions.

Teaching is integral to this practice. Over 25 years working at an art school, I have trained artists and accompanied projects across music, performance and the visual arts. My work also includes doctoral supervision across science, art and technology, and international workshops that open scientific ideas to people from other disciplines.

Sharing scientific thought

L.I.F.E. project, Performing Arts Forum, Saint-Erme, France. Initiated by Gabriel Catren.

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| 2022 | Deeply Organized Chaos
Workshop leader. Chaos and complexity explored through accessible concepts, fractals, sound experiments and embodied experience. 7–13 July. |
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| 2023 | Hot Identities and Vicious Abstractions
Assisted Gabriel Catren, with Jonathan Lorand, in a workshop on the foundations of mathematics. 21–29 October. |
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| 2025 | The Infinite, the Infinitesimal and the Imaginary
Assisted Gabriel Catren, with Jonathan Lorand and Emily Roff. An exploration of calculus, infinity and mathematical thought. 20–28 October. |
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Selected collaborations

Sound design and technological development in dialogue with the authors of these works.

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| 2023–2024 | Beehive / Colmena
Jorge Macchi
Multichannel sound realization. What the Night Tells the Day, PAC, Milan. |
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| 2023 | Glockenbuch IV (Spectre Maria dei Carmini)
Marcus Schmickler, in collaboration with Julian Rohrerhuber
Multichannel sound collaboration. Sound Microscopies programme, Biennale Musica, Venice. Credited by La Biennale. |
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| 2021 | Reliquias
Jorge Caterbetti
Technical collaboration on the installation dedicated to Julio López. Centro Cultural Kirchner, Buenos Aires. |
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| 2020 | La Cathédrale engloutie
Jorge Macchi
Development of the sound installation, with UNQ support. Musée cantonal des Beaux-Arts, Lausanne. 10 September–22 November. |
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| 2019 | Huella Sonora / Alfombra Sonora
Jorge Caterbetti
Technical collaboration on an interactive sound installation within Green Carpet. Centro Cultural de la Ciencia, Buenos Aires. |
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Selected CV

Artistic practice, scientific research and education are interdependent parts of my trajectory.

Research & teaching

2026–	Director, ArCiTeS Instituto de Investigaciones en Arte, Ciencia, Tecnología y Sociedad, Universidad Nacional de Quilmes.
2019–	Full Professor Escuela Universitaria de Artes, Universidad Nacional de Quilmes.
2017	Founder, Proyecto GRAPa Grupo de Relevamiento Acústico del Patrimonio.
2007	Founder, LAPSo Laboratory of Acoustics and Sound Perception, Universidad Nacional de Quilmes.
	CONICET researcher Physics, nonlinear dynamics, neuroscience and acoustics.
Director of Perspectiva Acústica, Etapa II (2025–2029), a UNQ research programme comprising seven projects.	

Education

2002	PhD in Physics Universidad de Buenos Aires.
1998	Licenciatura in Physics Universidad de Buenos Aires.

Exhibitions, residencies & support

2024	Sonic Crystal Room Festival de Nueva Ópera, opening programme, cheLA, Buenos Aires. With Oscar Edelstein.
2023	Transit · Artist residency & commission Northern Sustainable Futures, Moskosel, Sweden. With Mauro Zannoli, for iM Konsthall.
2022	Biocenosis · 13th Mercosul Biennial Porto Alegre, Brazil. Sigismond de Vajay & Kevin Lesquenner + LAPSo.
2021	Latin GRAMMY Cultural Foundation Research Grant With Proyecto GRAPa: recording canto con caja in its original acoustic environments.
2019	Sonic Crystal Room · First presentation Auditorio Nicolás Casullo, Universidad Nacional de Quilmes, Bernal.
2011	Lluvia de Arañas sobre el Riachuelo Collective project with Sigismond de Vajay and Buenos Aires Sonora. Noche de los Museos, Buenos Aires.

Research that informs the practice

These publications connect sonic crystals and acoustic time reversal with spatial perception, musical timbre and sound synthesis. They trace scientific questions that feed my artistic practice.

Current enquiry: vibrating matter

My recent research into acoustic time reversal extends into studio experiments with pulses travelling through steel sheets. I am exploring how bending a surface changes sound focusing, timbre and radiation, and how these changes can become bodily gestures and spatial listening experiences.

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meguia.github.io/dossier

vimeo.com/meguia

youtube.com/@manueleguia2915

Project photographs and stills supplied by Manuel Eguía. Authorship is credited on each project page. This selection draws on the artist's dossiers and CV, with institutional records used to verify dates and credits.

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| 2026 | The influence of visual context on auditory distance perception: A comparison between real and virtual visual environments
Cerviño, Lombera, Piceda, Eguía & Vergara · Applied Acoustics 250, 111334 |
| 2026 | Real-time sound synthesis from normal forms of dynamical systems
Ortega Brook, Ruiz & Eguía · Journal of New Music Research · Online publication 2026 |
| 2025 | Encrypted acoustical communication system using time reversal through a tunable sonic crystal
Gómez, Spiouzas & Eguía · The Journal of the Acoustical Society of America 158(3), 2100–2110 |
| 2022 | IRIS: A tunable sound sculpture based on acoustic periodic composites for musical performance
Salzano & Eguía · Journal of New Music Research 51, 27–39 |
| 2021 | Time reversal focusing in the audible range using a tunable sonic crystal
Gómez, Spiouzas & Eguía · The Journal of the Acoustical Society of America 149(6), 4024–4035 |
| 2017 | Auditory environmental context affects visual distance perception
Etchemendy, Abregú, Calcagno, Eguía, Vechiatti, Iasi & Vergara · Scientific Reports 7, 7189 |
| 2015 | An auditory illusion of proximity of the source induced by sonic crystals
Spiouzas, Etchemendy, Vergara, Calcagno & Eguía · PLOS ONE 10, e0133271 |
| 2014 | A comparative study of saxophone multiphonics: Musical, psychophysical and spectral analysis
Riera, Proscia & Eguía · Journal of New Music Research 43(2), 202–213 |
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